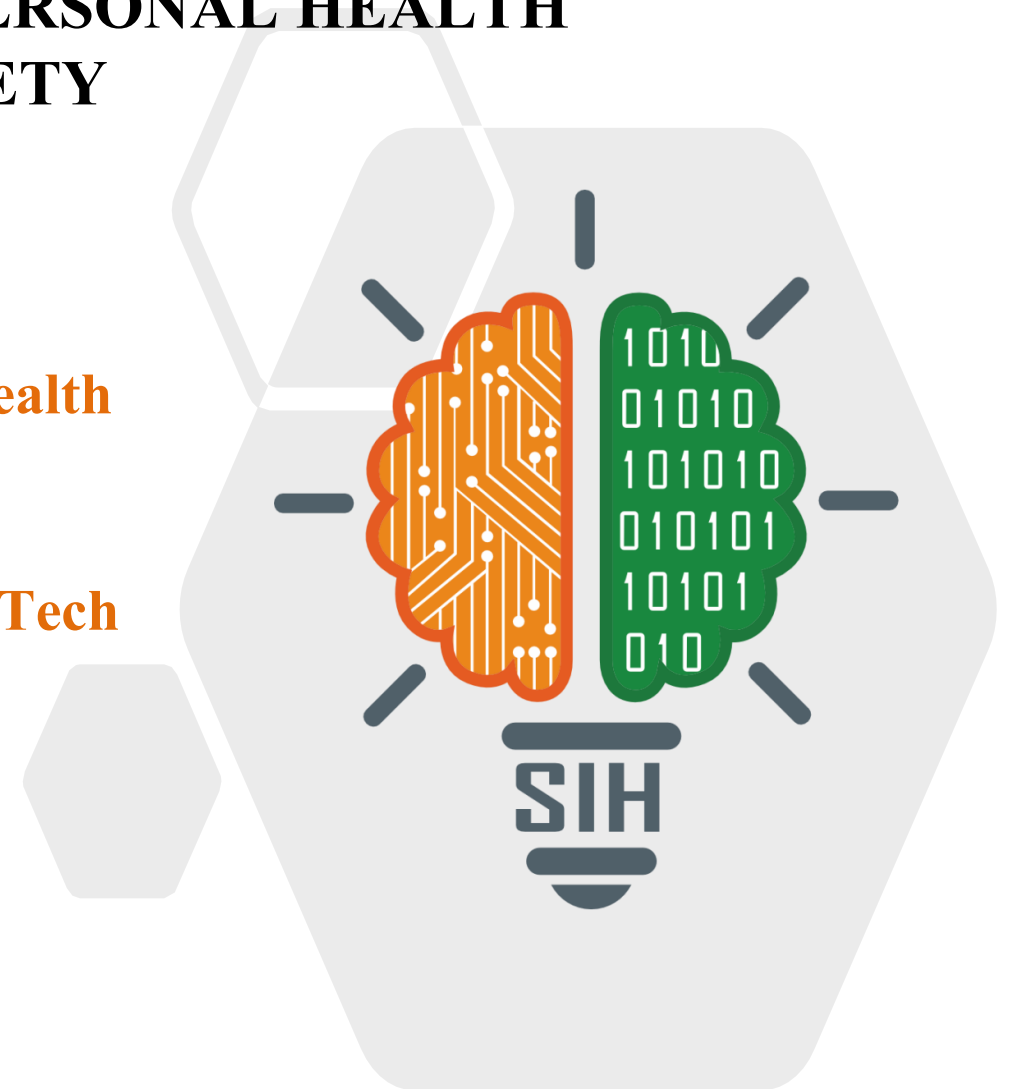


INTEGRATED AI HEALTH BELT FOR PERSONAL HEALTH MONITORING AND SAFETY

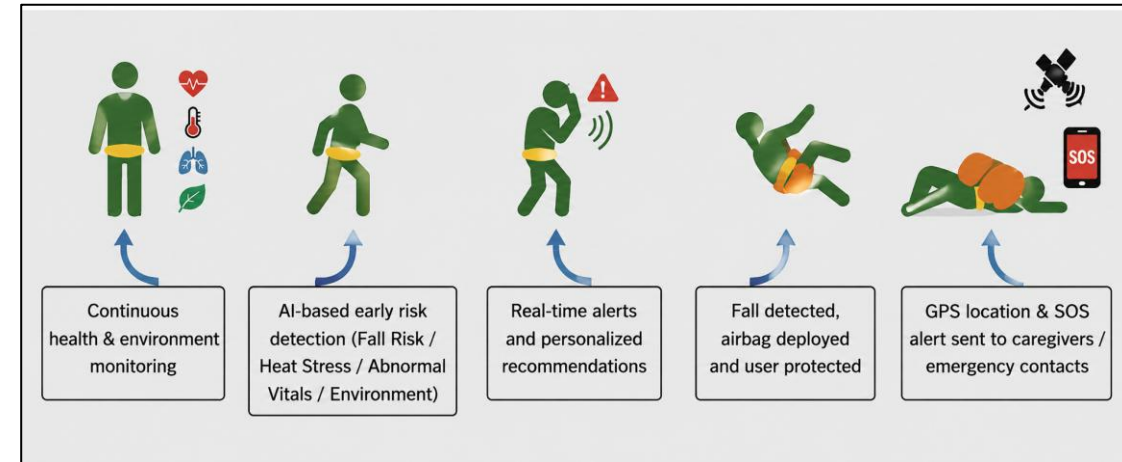
- Problem Statement ID - **SIH26181**
- Problem Statement Title - **Smart AI-Based Personal Health Guardian**
- Theme - **MedTech/BioTech/HealthTech**
- PS Category - **Hardware**
- Team ID - **-**
- Team Name - **Elite Crew**



INTEGRATED AI HEALTH BELT FOR PERSONAL HEALTH MONITORING AND SAFETY

PROPOSED SOLUTION

- A **smart wearable Personal Health Companion** designed for elderly individuals, rural populations, outdoor workers, and other vulnerable users. The belt continuously monitors **body movement, vital signs, and environmental conditions** using integrated sensors, with an **ESP32 enabling real-time edge processing**.
- An **Edge-AI health risk engine** analyzes sensor data to identify **fall risk, abnormal vital signs, fatigue, heat stress, and environmental health risks** at an early stage. It provides **real-time warnings and personalized recommendations**, while critical analysis can continue locally during network disruptions.
- When a **critical emergency is detected**, the system provides immediate protection through **automatic airbag deployment using a solenoid valve and CO₂ cartridge**, while **GPS-based SOS alerts** notify caregivers or emergency contacts. A connected health dashboard provides **risk scores, health trends, alerts, and recommendations**.



MOTIVE TO DO THIS PROJECT

The motive of this project is to develop a **smart wearable health companion** that continuously monitors the user's health and surroundings. It aims to **detect falls and potential health risks at an early stage** and provide real-time alerts. During critical situations, the system can provide **emergency assistance and automatic airbag protection**. Overall, the project focuses on improving **user safety, independence, and timely response to health emergencies**.

TECHNICAL STACK

mpu6050

edge AI

esp32

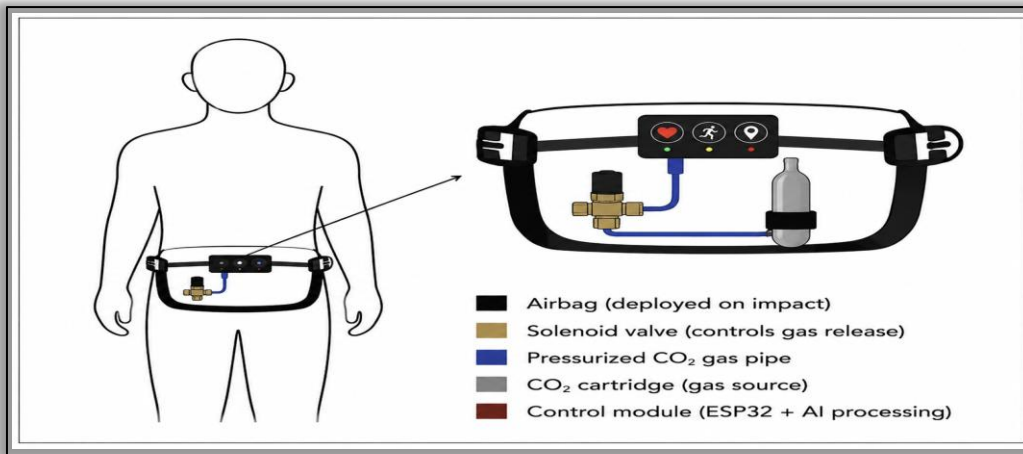
python or c

buzzer

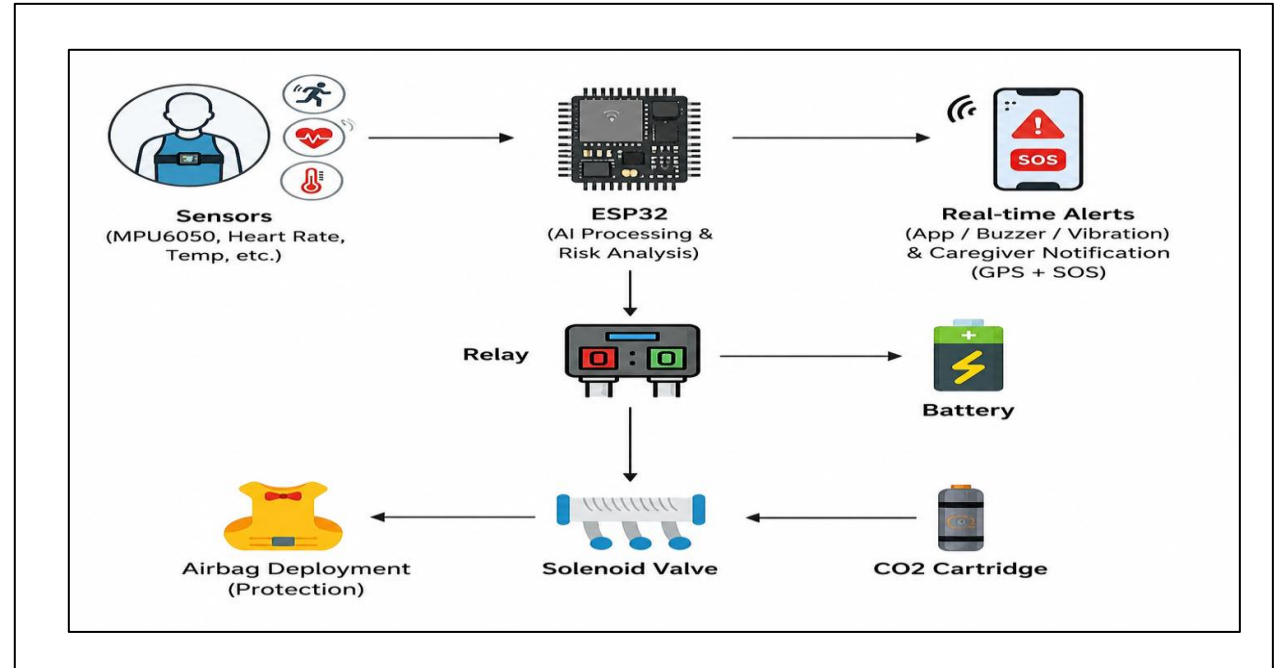
GPS

co2 gas

airbag



ARCHITECTURE FLOW



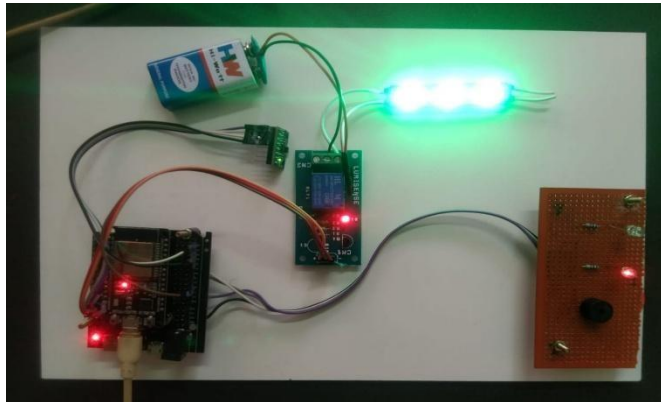
The system uses the **MPU6050 motion sensor** to monitor body movements and detect abnormal patterns that may indicate a fall. The data is processed by the **ESP32**, which performs AI-based risk analysis and provides real-time alerts. In critical cases, the ESP32 activates the **relay and solenoid valve** to release CO₂ and deploy the airbag, while **GPS and SOS alerts** notify caregivers or emergency contacts.

FEASIBILITY

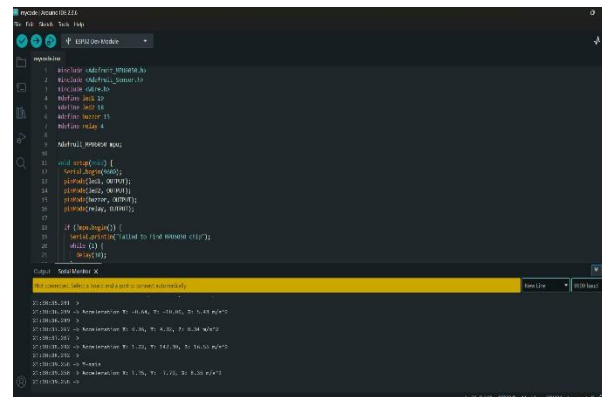
- Reliable Sensors
- Stable controller
- Lightweight design
- User friendly
- Easy daily use
- Low-cost production
- GPS & SOS Assistance

VIABILITY

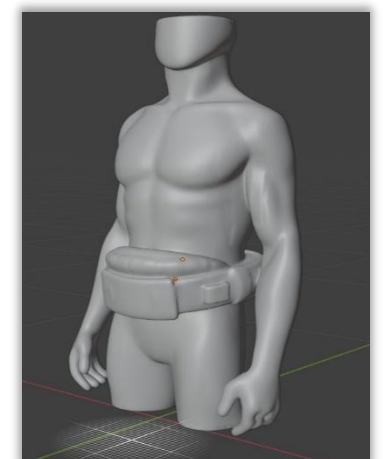
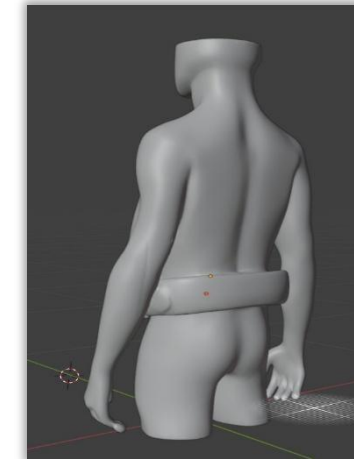
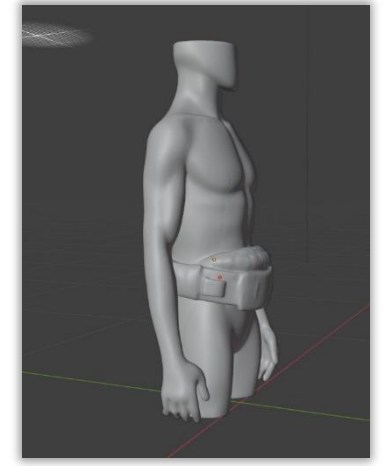
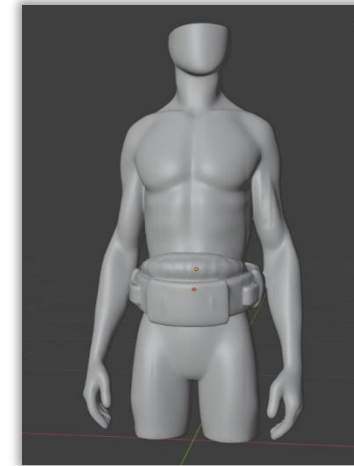
- Large market
- Rising Population
- Affordable pricing
- Wide adoption
- Improve safety
- Increase Independence
- Future upgrades



PROTOTYPE



CODE IMPLEMENTATION



IMPACT

Injury Reduction

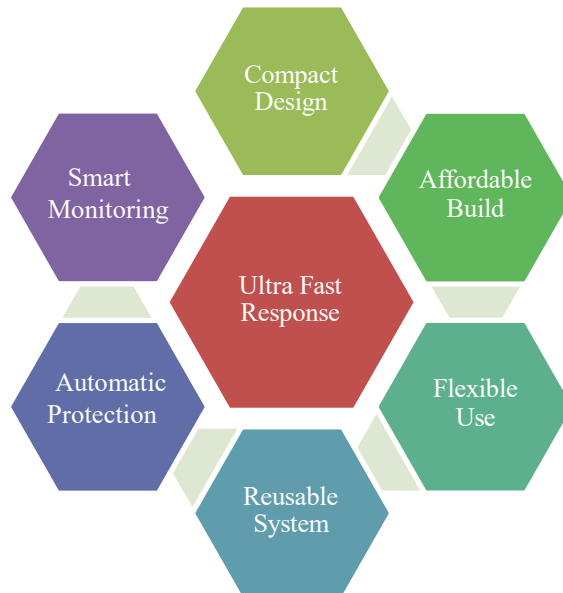
Early Risk
Detection

Improved User
Safety

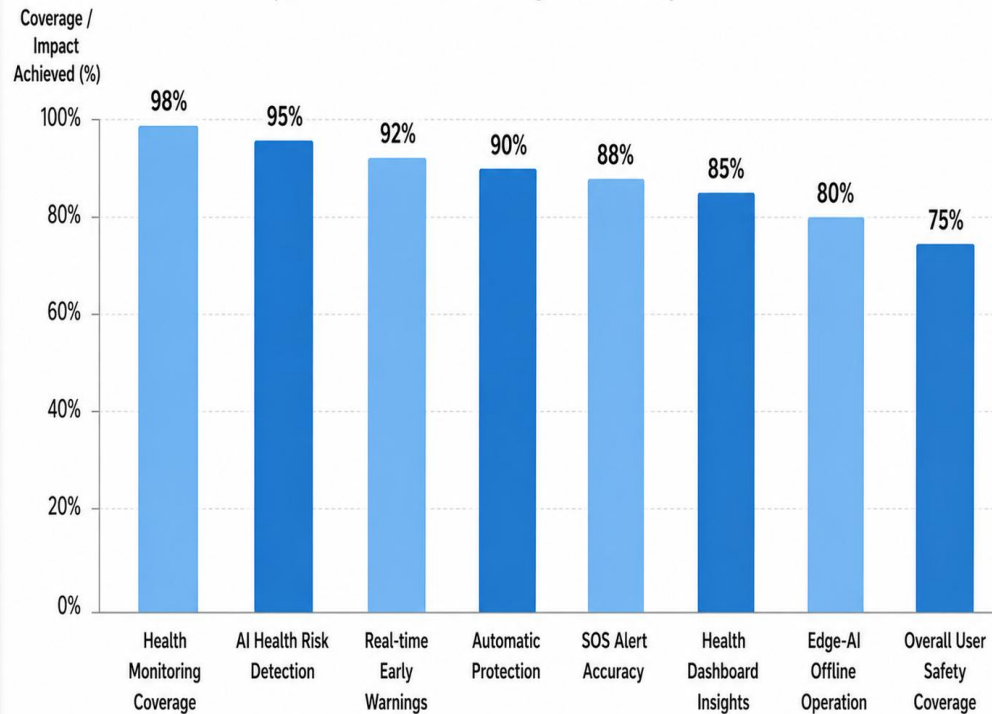
Emergency
Protection

Global Relevance

BENEFITS



**Impact of A Smart Wearable Personal Health Companion
Solution Across Key Health & Safety Dimensions**
(Expected Outcomes / Coverage Achieved by the Solution)



- <https://ieeexplore.ieee.org/document/10415018> - 2024
- <https://ieeexplore.ieee.org/document/10652375> - 2024
- <https://ieeexplore.ieee.org/document/11154594> - 2025
- <https://ieeexplore.ieee.org/document/11175568> - 2025
- <https://ieeexplore.ieee.org/document/10877492> - 2024